

Tab 2

Based on the current VPS state and container configuration, here is the complete port mapping for the Ivory infrastructure.

 VPS Port Mapping Table

Service	Internal Port	External/Host Port	Access Type	Description
Traefik	80, 443	80, 443	Public	Main Entrypoint (HTTP/HTTPS)
Traefik API	8080	8080	Public/Admin	Traefik Dashboard & API
VPS Bridge	3001	3001	Internal/Proxy	API Gateway (aivory-bridge)
n8n	5678	5678	Internal	Automation Engine
n8n-mcp	3000	3020	Local/Internal	MCP-n8n Bridge (Mapped to 3020)
Admin Dashboard	3300	3300	Internal	NextJS Console (avry-admin-dashboard)
Stag Frontend	80	N/A	Traefik \$\\rightarrow\$ 80	Staging Site (Routed via Traefik)
Redis	6379	6379	Localhost	Cache/Session Store
ZeroClaw	Dynamic	N/A	Internal	Backend Engine (Accessed via Bridge)

Key Routing Details

1. **Public Traffic Flow:** User $\rightarrow$ Port 443 $\rightarrow$ Traefik $\rightarrow$ Internal Container Port
2. **The Bridge Pattern:** The **VPS Bridge** (`aivory-bridge`) acts as the central dispatcher. Instead of exposing ZeroClaw or n8n directly to the internet, Traefik routes `api.aivory.id` to the Bridge on port 3001, which then forwards requests to the appropriate internal service.
3. **MCP-n8n Special Case:** The `n8n-mcp` container runs internally on 3000 but is mapped to the host's 3020 for specific local tool integrations.
4. **ZeroClaw Connectivity:** ZeroClaw typically operates within the `avry-admin-dashboard_app-network` and is reached by the VPS Bridge using the container name as the hostname, bypassing the need for a public host port.

Tab 1

🛠️ Component Detailed Mapping

1. Traefik (The Gateway)

- __Role__: Reverse Proxy & SSL Terminator.
- __Network__: `avry-admin-dashboard\_app-network`.
- __Configuration__:
 - __Entrypoints__: `web` (80), `websecure` (443).
 - __CertResolver__: `letsencrypt`.
 - __Routing Logic__: Uses Docker Labels to automatically discover services.
- __Key Route__: `Host(`stag.aivory.id`)` $\rightarrow$ `stag-frontend` container.

2. Stag Frontend (`stag-frontend`)

- __Role__: Staging environment for the public-facing website.
- __Tech Stack__: Nginx (Static HTML/JS/CSS).
- __Docker Config__:
 - __Volume__: `./frontend/frontend:/usr/share/nginx/html:ro`
 - __Labels__:
 - `traefik.http.routers.stag.rule=Host(`stag.aivory.id`)`
 - `traefik.http.routers.stag.entrypoints=websecure`
 - `traefik.http.services.stag.loadbalancer.server.port=80`

3. VPS Bridge (`vps-bridge`)

- __Role__: Secure API Gateway / Middleware.
- __Function__: Acts as a bridge between the frontend/console and the internal backend services (ZeroClaw/n8n). It handles authentication, request transformation, and routing to internal ports.
- __Key Logic__: `endpoints.js` and `server.js` define how external requests are mapped to internal service calls.

4. ZeroClaw (The Core)

- __Role__: The primary backend logic and execution engine.
- __Integration__: Communicates with the VPS Bridge and manages the state of the Ivory system.
- __Connectivity__: Accessible internally via the Docker network.

5. n8n (The Automation)

- __Role__: Workflow automation and integration engine.
- __Integration__: Triggered by ZeroClaw or the VPS Bridge to execute complex multi-step automations.

Configuration & Routing Flow

Request Lifecycle Example: `stag.aivory.id`

1. __User__ $\rightarrow$ Requests `https://stag.aivory.id`.
2. __Traefik__ $\rightarrow$ Matches `Host(`stag.aivory.id`)', terminates SSL, and forwards to `stag-frontend:80`.
3. __Nginx (stag-frontend)__ $\rightarrow$ Serves `index.html` from `/usr/share/nginx/html`.
4. __Browser__ $\rightarrow$ Executes `app.js`, which makes API calls to `api.aivory.id`.
5. __Traefik__ $\rightarrow$ Routes `api.aivory.id` to the __VPS Bridge__.
6. __VPS Bridge__ $\rightarrow$ Validates request $\rightarrow$ Forwards to __ZeroClaw__ or __n8n__.

Network Summary

- __External Network__: Public Internet $\rightarrow$ Traefik (Ports 80, 443).
- __Internal Network__: `avry-admin-dashboard\_app-network` (All containers communicate here via service names).

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Network Summary

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